

Dolly Pandya

Senior Software Development Engineer – Python | Test Automation | Embedded Systems

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Experienced **Software Development Engineer** with **8+ years** of hands-on expertise in developing and automating complex systems using **Python, C++, OOP** and **AI/ML frameworks**. Proven track record in **test automation, embedded system integration, memory profiling**, and **ECU validation** within the **automotive domain**. Skilled in designing **scalable automation scripts**, deploying **test pipelines**, and building **robust testing frameworks**. Adept at collaborating with **cross-functional teams**, implementing **MLOps best practices**, and driving **continuous integration** and **lab stability**. Strong knowledge of **object-oriented design (OOP)**, **version control**, and **Agile development**.

Core Competencies

Automation Framework Development, Python Scripting, OOP, ECU Integration Testing, MLOps Concepts (Versioning & Monitoring), CI/CD Pipeline Implementation, AWS, Test Strategy & Execution, Memory Profiling & Performance Analysis, Functional Safety Validation, REST API Design, Cross-Functional Collaboration, Agile/Scrum Methodology, Defect Triage & Reporting, Offshore Team Coordination, Data Logging & Pipeline Development, GitHub Workflow Management

Technical Skills

Python, C++, OOP, FastAPI, JSON, XML, CANalyzer, Appium, Jenkins, Git, Jira, Linux, Wireshark, Squish, PyCharm, Eclipse, Visual Studio, Selenium, Slash Framework, Google Test, Google Mock, Azure, SQL, GitHub Copilot, Golang, AWS, Provtech/XML

Experience

Software Development Engineer

Company: **Ford Motor Company**

Dearborn, MI

June 2017 - Present

Developed and maintained advanced automation frameworks for **SYNC** and **Phoenix infotainment systems** using **Python** and **C++, OOP** supporting functional validation, ECU integration, and memory profiling. Led automation efforts across **SYNC3, SYNC4, Phoenix** and **Menlo platforms**, delivering **70+ test scripts** and managing **nightly pipelines with 90+ steps**. Collaborated with **cross-functional and offshore teams** to enhance **lab stability**, implement **CI/CD workflows**, and align testing strategies with evolving system requirements.

- Developed **automated test frameworks** and Python-based scripts for Ford's infotainment systems (**SYNC3, SYNC4, Menlo, Phoenix**), improving test efficiency and coverage.
- Authored **70+ reusable automation scripts and utilities**, reducing manual testing time by **30%** and improving regression reliability.
- Created a **2,300+ line High Water Mark test suite** to log memory usage during **5,000+ HMI transitions**, enabling early memory leak detection.
- Built **nightly pipelines with 93 test steps**, capturing recurring crashes and black screens—boosting product reliability and defect traceability.
- Designed diagnostic verification scripts using **CANalyzer, packet capture (pcap) analysis**, and JSON updates to support **Functional Safety testing**.
- Streamlined triage and dashboard publishing through **TestGuide integration**, reducing resolution times and increasing transparency across teams.
- Applied **MLOps principles** to automation scripts for **version control, monitoring**, and **scalable deployment** aligned with lab execution cycles.
- Contributed to **CI/CD pipeline setup** using **Jenkins** and **Git**, managed **branching strategies** and **pull requests**, participated in **Agile ceremonies** (sprint planning, grooming, code reviews), **mentored junior engineers**, and piloted tools like **GitHub Copilot** to enhance **team productivity** and **deployment efficiency**.

Education

MS in Computer Science,

Oakland University, Rochester, MI

BS in Electronics and Communication,

S.S Engineering College, Gujarat, India

U.S Citizen